

Geometry learned on training faces predicts unseen faces

One train-to-test question

Unseen face groups land much closer than chance

1

TRAIN

Choose 3 attributes and learn their 8 corner locations.

2

STOP LEARNING

Do not change the directions or corner predictions using test data.

3

TEST ON UNSEEN IMAGES

Group 4,000 unseen images into 8 groups. Compare their centers with the predictions.

Test labels form the 8 groups; they never refit the geometry.

VICReg

actual labels
0.20

shuffled labels
about 1.00

5.0x lower error

I-JEPA

actual labels
0.25

4.0x lower error

0 of 5,000 shuffles
matched this well
 $p = 0.0002$ (both models)

0.00

0.25

0.50

0.75

1.00

corner mismatch

0 = perfect match

1 = shuffled-label baseline